

Software Requirements Specification (SRS)

Project: Blood Bank Management System (BBMS) — Web Application with Role-Based Access Control, Inventory Tracking, and Hospital Blood Request Management (Node.js / Express / SQLite / Tailwind CSS)

Version: 1.0

Authors: Uttam (PES1UG24CS697), Abhinav K (PES1UG24CS701), Akshay Arcot (PES1UG24CS705), Navaneeth Tanuboddi (PES1UG24CS709) — Team T9

Date: 14-09-2026

Status: Draft for review

Revision history

Version	Date	Author	Change summary	Approval
0.1	09-09-2026	Team T9	Initial draft: project scope, user roles, core entities	Approved
0.5	11-09-2026	Team T9	Added functional requirements (FRs), inventory rules, and API specifications	Approved
1.0	14-09-2026	Team T9	Complete SRS with security requirements, UML use-case diagrams, test cases, and RTM	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	Prof. Dept of CSE	coordinator.cse@pes.edu	14-09-2026
Course Instructor	Faculty Advisor	instructor.cse@pes.edu	14-09-2026
Team Lead	Navaneeth Tanuboddi	navaneeth.tanuboddi@gmail.com	14-09-2026

Table of Contents

- 1. [Introduction](#1-introduction)
- 2. [Overall description](#2-overall-description)
- 3. [External interface requirements](#3-external-interface-requirements)
- 4. [System features (detailed)](#4-system-features-detailed)
- 5. [Non-functional requirements (detailed)](#5-non-functional-requirements-detailed)
- 6. [Quality attributes & Acceptance tests](#6-quality-attributes--acceptance-tests)
- 7. [System models and diagrams (UML use-case)](#7-system-models-and-diagrams)
- 8. [Requirements Traceability Matrix (RTM)](#8-requirements-traceability-matrix-rtm)

1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Blood Bank Management System (BBMS), an enterprise-grade, role-based full-stack web application developed using Node.js, Express, SQLite, and Tailwind CSS. It defines the functional and non-functional requirements, external interfaces, security objectives and controls, system architectural models, and the verification criteria against which the delivered system will be assessed. The intended readers are the developer team (Team T9), reviewing faculty instructor, course coordinator, and laboratory evaluators.

1.2 Scope

The Blood Bank Management System encompasses the end-to-end operational lifecycle of voluntary blood donation, blood unit testing and processing, inventory management, hospital blood requisition, and emergency dispatch. The system provides role-based web portals for Donors, Hospital Representatives, Blood Bank Staff, and System Administrators.

Key in-scope subsystems include: donor pre-screening health questionnaires, appointment scheduling, digital donor cards, blood unit collection tracking with unique Bag IDs, mandatory serological infection screening (HIV, Hepatitis B/C, Syphilis, Malaria),

component separation (PRBC, Platelets, Fresh Frozen Plasma), First-Expired-First-Out (FEFO) stock monitoring, hospital requisition submission with clinical urgency prioritization, automated ABO/Rh cross-matching verification, cold-chain dispatch handover manifests, immutable audit logging, and a public availability dashboard.

Explicitly out of scope: Direct hardware automation of centrifuges/refrigeration IoT sensors, online monetary payment gateways (blood is donated voluntarily without commercial sale), and direct clinical transfusion administration inside operating theaters.

1.3 Audience

Developer Team (Team T9), Software Quality Assurance / Test Engineers, Course Instructors, Academic Evaluators, and Hospital Blood Transfusion Committees.

1.4 Definitions and Acronyms

Term	Meaning
BBMS	Blood Bank Management System — integrated web platform for blood lifecycle tracking.
RBAC	Role-Based Access Control — security architecture restricting access based on user role.
PRBC	Packed Red Blood Cells — red cell component used to treat acute hemorrhage and severe anemia.
FFP	Fresh Frozen Plasma — plasma frozen within 8 hours, containing labile coagulation factors.
Platelet Concentrate	Thrombocytes harvested to prevent bleeding in thrombocytopenic patients (5-day shelf life).
Serology Screening	Laboratory diagnostic assays for transfusion-transmissible infections (TTIs).
Quarantine	Holding status isolating blood units pending test results or discarding non-conforming units.
Cross-Matching	Compatibility testing between donor erythrocytes and recipient serum prior to issue.
FEFO	First-Expired-First-Out — inventory distribution strategy prioritizing closest expiration date.
RTM	Requirements Traceability Matrix — mapping requirements to modules, test cases, and status.

2. Overall description

2.1 Product perspective

The Blood Bank Management System operates as a centralized, self-contained client-server web application. The backend is powered by Node.js and Express RESTful services backed by an ACID-compliant SQLite relational database running in Write-Ahead Logging (WAL) mode for high concurrent read throughput. The frontend utilizes responsive semantic HTML5, vanilla modern JavaScript (ES6+), and Tailwind CSS for mobile-friendly UI rendering. The application is completely containerizable and can run locally or behind reverse proxies (Nginx, Cloudflare Tunnels) for secure HTTPS access.

2.2 Major product functions

- Donor registration, medical eligibility self-screening, and digital donor card generation.
- Blood donation appointment booking and blood drive camp scheduling.
- Blood collection logging with unique barcode-compatible Bag IDs and donor linkage.
- Laboratory serology testing entry with automated quarantine enforcement for reactive units.
- Blood component separation tracking (PRBC, Platelets, FFP) with distinct temperature and shelf lives.
- Real-time blood stock inventory management categorized by 8 blood groups and component types.
- First-Expired-First-Out (FEFO) inventory allocation and proactive expiration threshold alerts.
- Hospital blood requisition workflow supporting Emergency/STAT and routine priority queues.
- Automated ABO/Rh compatibility cross-matching engine.
- Cryptographically signed dispatch manifests and chain-of-custody delivery verification.
- Immutable system audit logging and regulatory compliance reporting.

2.3 User roles and characteristics

- **Public Donor** — Voluntary blood donor. Expects intuitive self-service portal, eligibility guidance, simple appointment booking, and instant access to digital donor cards.

- **Hospital Representative** — Authorized hospital physician or blood bank coordinator. Submits urgent blood requisitions, tracks fulfillment status, and confirms custody transfer.
- **Blood Bank Staff / Lab Technician** — Laboratory professional logging blood collections, recording infectious disease screening assays, separating components, and managing stock allocations.
- **System Administrator** — IT/Operations lead managing user roles, configuring blood bank parameters, inspecting audit trails, and generating regulatory compliance reports.
- **Course Evaluator / Inspector** — Academic evaluator reviewing source code modularity, test coverage, relational schemas, and adherence to Software Engineering standards.

2.4 Operating environment

- **Server:** Node.js 18+ runtime on macOS (Apple Silicon / Intel), Linux (Ubuntu 22.04 LTS), or Windows 10/11 Server.
- **Database:** SQLite 3.x embedded database with WAL mode enabled; zero external database daemon required.
- **Client:** Modern evergreen web browsers (Chrome 100+, Safari 15+, Firefox 100+, Edge 100+) on desktop, tablet, and mobile devices.
- **Network:** Standard HTTP/1.1 and HTTP/2 over TCP ports 3000 / 443 with TLS encryption.

2.5 Constraints and assumptions

- Standards compliance: Clean architectural separation between REST controllers, business service logic, and database access models.
- Relational integrity: Foreign keys strictly enforced across all database tables (donors, units, requests, logs).
- Safe dispensing: The system strictly blocks allocation of expired, un-tested, or serologically reactive blood units.
- Physical pre-condition: Clinical vitals (hemoglobin ≥ 12.5 g/dL, blood pressure, weight ≥ 45 kg) are verified by physical nursing staff at collection.
- Regulatory compliance: Complete audit logging of all inventory changes to fulfill National Blood Transfusion Council guidelines.

3. External interface requirements

3.1 User interfaces

The web interface is engineered with responsive, accessible Tailwind CSS components. Key screens include:

- **Public Landing & Availability Portal:** Real-time stock summary, donor educational guidelines, and blood camp schedules.
- **Donor Portal:** Clean health questionnaire, appointment selector with time-slot reservation, and printable digital donor card.
- **Staff Operations Dashboard:** Rapid unit logging with Bag ID scanner support, serology results checklist, and quarantine status toggles.
- **Hospital Order Portal:** Structured requisition form with blood group selectors, unit counts, priority indicators, and delivery tracking.
- **Admin Control Center:** Interactive stock analytics, user role administration, and immutable audit trail tables.

3.2 Hardware Interfaces

- **Client Workstation / Mobile Device:** Minimum 1024x768 resolution for staff dashboard; responsive down to 360px width for donor portal.
- **Barcode / QR Scanner:** Optional standard USB/Bluetooth HID keyboard emulation for rapid Bag ID and Donor Card scanning.
- **Host Server:** Standard computing hardware (minimum 1 vCPU, 1 GB RAM, 10 GB storage).

3.3 Software Interfaces

- **Operating System:** macOS, Linux (Debian/Ubuntu/CentOS), or Windows 10/11.
- **Runtime Environment:** Node.js (v18.x or v20.x LTS).
- **Database Engine:** SQLite3 embedded database engine.
- **Tunnel / Proxy:** Cloudflare Tunnel daemon / Nginx for reverse proxying and automated SSL/TLS termination.

3.4 Communications Interfaces

RESTful JSON APIs over HTTP/HTTPS. Stateful session management via HTTP-only, SameSite secure cookies and JWT tokens.

4. System features (detailed)

Each requirement below carries acceptance criteria and a reference test case. Functional requirement IDs follow the pattern BBMS-F.

4.1 User Authentication and Role Management

Description: Establish identity, enforce role-based access control (RBAC), and protect user sessions.

Req ID	Requirement (shall...)	Type	Prio.	Source / Stakeholder	Acceptance criteria / Test ref	Dependencies
BBMS-F-001	The system shall authenticate users via email and bcrypt-hashed passwords, generating secure session tokens.	Functional	High	Security / Admin	AC: Valid credentials grant access; invalid credentials return 401 Unauthorized. Test: TC-AUTH-01	User database
BBMS-F-002	The system shall enforce RBAC across Donor, Hospital, Staff, and Admin roles, returning HTTP 403 for unauthorized routes.	Functional	High	Security / Evaluator	AC: Donors cannot access staff endpoints; hospitals cannot alter inventory. Test: TC-AUTH-02	BBMS-F-001
BBMS-F-003	The system shall temporarily lock out accounts after 5 consecutive failed login attempts within 15 minutes.	Functional	Medium	Security	AC: 5th failure triggers 15-min lockout; event logged to audit table. Test: TC-AUTH-03	BBMS-F-001
BBMS-F-004	The system shall provide a password reset flow utilizing time-limited (15 min) cryptographic tokens.	Functional	Medium	Donor / Hospital	AC: Expired or reused tokens are rejected with error diagnostic. Test: TC-AUTH-04	BBMS-F-001
BBMS-F-005	The system shall terminate sessions and invalidate authentication tokens upon explicit user logout.	Functional	High	User / Security	AC: Subsequent requests with cleared session return 401 Unauthorized. Test: TC-AUTH-05	BBMS-F-001

4.2 Donor Management and Appointment Scheduling

Description: Facilitate donor registration, medical eligibility self-screening, and appointment management.

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria / Test Case	Dependencies / Comments
BBMS-F-006	The system shall capture donor profile details (name, DOB, blood group, contact, emergency contact).	Functional	High	AC: Profile stored with unique Donor ID; duplicate emails rejected. Test: TC-DNR-01	Donor portal
BBMS-F-007	The system shall present a mandatory clinical eligibility questionnaire (age, weight, hemoglobin, travel).	Functional	High	AC: Ineligible responses flag donor and block immediate appointment. Test: TC-DNR-02	BBMS-F-006
BBMS-F-008	The system shall enforce a minimum 90-day interval between whole blood donation appointments.	Functional	High	AC: Appointment booking disabled if prior donation < 90 days ago. Test: TC-DNR-03	Donation history
BBMS-F-009	The system shall generate a digital donor card displaying donor photo placeholder, blood group, and QR code.	Functional	Medium	AC: Digital card renders correctly and QR encodes valid donor verification URL. Test: TC-DNR-04	BBMS-F-006
BBMS-F-010	The system shall allow donors to schedule, view, and cancel upcoming appointments up to 24 hours prior.	Functional	Medium	AC: Slot reservations update camp capacity counts in real time. Test: TC-DNR-05	BBMS-F-008

4.3 Blood Collection, Testing, and Inventory Processing

Description: Record collected blood, enforce serology testing, quarantine infectious units, and separate components.

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria / Test Case	Dependencies / Comments
--------	------------------------	------	----------	---------------------------------	-------------------------

BBMS-F-011	The system shall record blood collections with unique Bag IDs, collection timestamps, and phlebotomist IDs.	Functional	High	AC: Unique alphanumeric Bag ID assigned; status initialized to PENDING_TESTING. Test: TC-INV-01	Staff auth
BBMS-F-012	The system shall require mandatory serology test results (HIV, HBV, HCV, Syphilis, Malaria) before clearance.	Functional	High	AC: Units cannot be marked AVAILABLE without all 5 negative test results. Test: TC-INV-02	BBMS-F-011
BBMS-F-013	The system shall automatically quarantine and permanently lock any blood unit testing positive for any TTI.	Functional	High	AC: Reactive unit status set to QUARANTINED; allocation disabled permanently. Test: TC-INV-03	BBMS-F-012
BBMS-F-014	The system shall track component separation into PRBC (35d shelf life), Platelets (5d), and FFP (365d).	Functional	High	AC: Child unit records created with accurate expiry dates and storage temperatures. Test: TC-INV-04	BBMS-F-012
BBMS-F-015	The system shall enforce FEFO (First-Expired-First-Out) stock ranking and trigger low/expiring inventory alerts.	Functional	Medium	AC: Inventory listings display closest expiry first; expiring units highlighted in red. Test: TC-INV-05	BBMS-F-014

4.4 Hospital Blood Requisition and Fulfillment

Description: Process hospital requisitions, verify compatibility, reserve units, and track dispatch handover.

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria / Test Case	Dependencies / Comments
BBMS-F-016	The system shall allow verified hospitals to submit blood requests with blood group, units, patient ID, and priority.	Functional	High	AC: Requisition recorded with status PENDING; emergency orders flagged STAT. Test: TC-REQ-01	Hospital auth
BBMS-F-017	The system shall verify ABO/Rh compatibility rules and check real-time stock availability for requisitions.	Functional	High	AC: Incompatible blood groups prevented; universal donor (O-) permitted when configured. Test: TC-REQ-02	BBMS-F-015
BBMS-F-018	The system shall allow staff to approve requisitions and transition corresponding units to RESERVED status.	Functional	High	AC: Approved units removed from available stock count immediately. Test: TC-REQ-03	BBMS-F-017
BBMS-F-019	The system shall generate a dispatch manifest containing unit Bag IDs, issuing officer, and hospital signature lines.	Functional	High	AC: Printable dispatch manifest generated with cold-chain verification checklist. Test: TC-REQ-04	BBMS-F-018
BBMS-F-020	The system shall record delivery confirmation and update request status to FULFILLED upon hospital receipt.	Functional	High	AC: Receiving officer name and timestamp recorded; request marked FULFILLED. Test: TC-REQ-05	BBMS-F-019

4.5 Persistence, Audit Logging, and Reporting

Description: Maintain immutable operational audit trails, generate regulatory reports, and provide public availability data.

Req ID	Requirement (shall...)	Type	Prio.	Source / Stakeholder	Acceptance criteria / Test ref	Dependencies
BBMS-F-021	The system shall record all unit status changes, approvals, and discards in an append-only audit trail table.	Functional	High	Regulatory / Admin	AC: Audit record contains ISO 8601 timestamp, user ID, IP, and state diff. Test: TC-AUD-01	SQLite audit table
BBMS-F-022	The system shall present an interactive analytics dashboard of current stock, pending requests, and collection stats.	Functional	Medium	Admin / Staff	AC: Dashboard metrics match underlying database counts precisely. Test: TC-AUD-02	BBMS-F-015
BBMS-F-023	The system shall allow administrators to export inventory and requisition reports in CSV and printable formats.	Functional	Medium	Admin	AC: Generated CSV contains valid headers and sanitized data. Test: TC-AUD-03	BBMS-F-021
BBMS-F-024	The system shall provide an unauthenticated public portal displaying real-time aggregate blood availability.	Functional	Medium	Public / Donor	AC: Public view displays stock counts without exposing sensitive donor records. Test: TC-AUD-04	BBMS-F-015

BBMS-F-025	The system shall support automated database backups and integrity verification without process disruption.	Functional	High	Sysadmin / Dev	AC: SQLite WAL backup executes cleanly; PRAGMA integrity_check returns ok. Test: TC-AUD-05	SQLite engine
-------------------	--	------------	------	----------------	--	---------------

5. Non-functional requirements (detailed)

NFRs are measurable quality constraints and are tied to verification activities.

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
BBMS-NF-001	Under a simulated load of 50 concurrent users, the application shall respond to 95% of HTTP API requests within 250 ms.	Performance	High	ApacheBench / k6 benchmark demonstrates p95 < 250 ms over 1,000 requests. Test: TC-NF-PERF-01
BBMS-NF-002	The system shall maintain 99.9% availability during operational hours with zero unhandled exceptions in 72h stress tests.	Reliability	High	Continuous automated API test harness runs for 72 hours without server crash. Test: TC-NF-REL-01
BBMS-NF-003	All user interfaces shall be responsive across viewports from 360px to 1920px and adhere to WCAG 2.1 Level AA color contrast.	Usability	Medium	Lighthouse Accessibility score ≥ 95; zero layout overflow on mobile breakpoints. Test: TC-NF-UX-01
BBMS-NF-004	The backend application and database shall deploy and run on macOS, Ubuntu 22.04 LTS, and Windows 10/11 without modification.	Portability	Medium	Automated clean install and startup test succeeds on all 3 target OS platforms. Test: TC-NF-PORT-01
BBMS-NF-005	The codebase shall maintain modular separation between routes, controllers, models, and public assets with full API docs.	Maintainability	Medium	Code inspection confirms zero raw SQL in route handlers; ESLint passes with 0 errors. Test: TC-NF-MNT-01

5.1 Security

5.1.1 Security objectives

The Blood Bank Management System processes sensitive medical information, confidential donor contact details, and critical hospital blood inventory. The system threat model addresses both external malicious actors (unauthorized web access, data tampering) and internal threats (unauthorized role escalation or accidental release of quarantined blood).

- **SO-1 (Confidentiality of Donor Health Information):** Donor health screening questionnaires, serology results, and personal identifiable information (PII) shall be visible only to authorized medical staff and the donor themselves.
- **SO-2 (Integrity of Blood Inventory and Quarantine):** Blood unit safety statuses (Quarantined, Available, Expired) shall only be modifiable through authenticated, logged transactions. No expired or infected unit shall ever be issued.
- **SO-3 (Account Authentication & Credential Protection):** User passwords shall be hashed using salted bcrypt with minimum 10 rounds; plaintext passwords shall never appear in logs or responses.
- **SO-4 (Audit Non-Repudiation):** All inventory allocations, status transitions, and hospital dispatches shall generate immutable audit trail records.

Stated limitation: Server operating system-level physical security and file permissions are assumed to be managed by the system administrator. Database file tampering directly on disk bypasses application-level controls; SQLite file permissions must be restricted to the node process owner (0600 on POSIX).

5.1.2 Security requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
BBMS-SR-001	All user passwords shall be stored only as salted bcrypt hashes with at least 10 salt rounds; plaintext passwords shall never be logged or echoed.	Security	High	Database inspection reveals only \$2b\$ hashes; no plaintext password in server logs. Test: TC-SEC-01
BBMS-SR-002	All SQL queries shall utilize parameterized prepared statements; dynamic string concatenation in SQL queries shall not appear in the codebase.	Security	High	Code review and sqlmap injection scan confirm zero SQL injection vulnerabilities. Test: TC-SEC-02
BBMS-SR-003	Session authentication tokens and cookies shall use HttpOnly, SameSite=Strict, and Secure flags to prevent XSS session hijacking.	Security	High	Browser DevTools inspection verifies HttpOnly and SameSite cookie attributes. Test: TC-SEC-03

BBMS-SR-004	Role authorization middleware shall protect every non-public API endpoint, returning HTTP 403 for unauthorized privilege escalation attempts.	Security	High	Automated privilege escalation tests verify that donor accounts cannot access /api/staff or /api/admin. Test: TC-SEC-04
BBMS-SR-005	All user inputs shall be validated against strict schemas (whitelisted characters, length limits, email regex) before backend processing.	Security	High	Malformed payloads and cross-site scripting (XSS) probe strings are rejected with 400 Bad Request. Test: TC-SEC-05
BBMS-SR-006	Authentication endpoints shall enforce rate limiting (max 10 requests per minute per IP) to mitigate automated brute-force attacks.	Security	Medium	Rapid successive login attempts receive HTTP 429 Too Many Requests response. Test: TC-SEC-06

6. Quality attributes & acceptance tests

Exit criteria for acceptance:

- Every high-priority functional requirement (BBMS-F-001 through BBMS-F-025) is fully implemented and passes all automated verification test suites.
- No non-functional requirement fails: in particular, the automated 72-hour stress test and response latency benchmarks (< 250 ms p95) are blocking.
- All six security requirements (BBMS-SR-001 to BBMS-SR-006) pass with zero critical or high vulnerabilities identified during static analysis and dynamic penetration testing.
- The Requirements Traceability Matrix (RTM) in Section 8 demonstrates 100% bidirectional coverage with all test cases evaluated with status A (Accepted).

Acceptance test suites: TC-AUTH (authentication and session lifecycle), TC-DNR (donor profile, questionnaire, and appointment scheduling), TC-INV (blood collection, serology testing, component separation, and FEFO stock logic), TC-REQ (hospital requisitions, cross-matching, allocation, and dispatch), TC-AUD (audit trail persistence and reporting), TC-NF (performance, reliability, usability, and portability), TC-SEC (password security, SQL injection, XSS, RBAC enforcement, and rate limiting).

Verification methods used: Automated end-to-end API integration tests using Node.js test runner and Supertest; unit tests verifying cross-matching logic; load testing via ApacheBench; accessibility audits using Lighthouse; manual walkthroughs of all role-based UI dashboards.

7. System models and diagrams

7.1 Use-case diagram — Donor and Hospital Portal Workflows

Figure 7.1 illustrates the external-facing operational boundary of the Blood Bank Management System. The primary external actors are the Voluntary Donor and the Hospital Representative. The Donor interacts with the public portal to register an account, complete the pre-donation medical eligibility questionnaire, schedule or reschedule donation appointments, view past donation history, and access their digital donor card. The appointment scheduling use case automatically includes eligibility verification. The Hospital Representative authenticates via a verified institutional account to submit routine or emergency (STAT) blood requisitions, track real-time fulfillment status, and acknowledge blood unit delivery upon arrival.

[image]

Figure 7.1 — Use-case diagram: Donor and Hospital Portal Workflows

7.2 Use-case diagram — Blood Bank Staff & Administrator Workflows

Figure 7.2 illustrates the internal operations and administrative boundary. The primary actors are the Blood Bank Staff / Lab Technician and the System Administrator. Staff record incoming physical blood collections with unique Bag IDs, log serological diagnostic assay results, flag and discard contaminated or reactive units via quarantine controls, separate whole blood into red cell, platelet, and plasma components, and dispatch allocated blood to hospitals. System Administrators authenticate with elevated credentials to manage user accounts, assign role permissions, configure inventory threshold alerts, review immutable system audit trails, and generate statutory blood supply reports.

[image]

Figure 7.2 — Use-case diagram: Blood Bank Staff & Administrator Workflows

8. Requirements Traceability Matrix (RTM)

Status legend: N = Not Run, P = Pass, A = Accepted / Approved.

Req ID	Requirement Short Name	Section Ref	Module	Test Case(s)	Status (N/P/A)	Comments
BBMS-F-001	User authentication	4.1	Auth / Session	TC-AUTH-01	A	bcrypt hashing & session tokens verified
BBMS-F-002	Role-Based Access Control	4.1	Auth / RBAC	TC-AUTH-02	A	Donor/Hospital/Staff/Admin permissions enforced
BBMS-F-003	Failed login lockout	4.1	Auth / Security	TC-AUTH-03	A	Lockout after 5 failed attempts verified
BBMS-F-004	Password reset flow	4.1	Auth / Recovery	TC-AUTH-04	A	15-min cryptographic token reset verified
BBMS-F-005	Session logout termination	4.1	Auth / Session	TC-AUTH-05	A	Immediate cookie/token invalidation verified
BBMS-F-006	Donor profile capture	4.2	Donor Module	TC-DNR-01	A	Unique donor record and blood group stored
BBMS-F-007	Medical eligibility screening	4.2	Donor Screening	TC-DNR-02	A	Clinical eligibility questionnaire enforced
BBMS-F-008	90-day donation interval	4.2	Donor Scheduling	TC-DNR-03	A	Interval check prevents premature booking
BBMS-F-009	Digital donor card generation	4.2	Donor Portal	TC-DNR-04	A	Digital card with QR code verified
BBMS-F-010	Donation history & cancel	4.2	Donor Portal	TC-DNR-05	A	Donation history and appointment cancel verified
BBMS-F-011	Blood collection logging	4.3	Collection / Lab	TC-INV-01	A	Unique alphanumeric Bag IDs generated
BBMS-F-012	Serology screening tests	4.3	Testing / Lab	TC-INV-02	A	Mandatory 5-infection testing verified
BBMS-F-013	Quarantine reactive units	4.3	Lab / Quarantine	TC-INV-03	A	Reactive units locked and prevented from issue
BBMS-F-014	Component separation	4.3	Inventory Engine	TC-INV-04	A	PRBC, Platelets, and FFP shelf lives enforced
BBMS-F-015	FEFO stock management	4.3	Inventory Engine	TC-INV-05	A	First-expired-first-out prioritization verified
BBMS-F-016	Hospital requisition submit	4.4	Hospital Portal	TC-REQ-01	A	Routine and STAT blood requests processed
BBMS-F-017	ABO/Rh cross-matching	4.4	Matching Engine	TC-REQ-02	A	Cross-match compatibility matrix enforced
BBMS-F-018	Request review & reserve	4.4	Staff / Inventory	TC-REQ-03	A	Inventory reservation upon approval verified
BBMS-F-019	Dispatch manifest generate	4.4	Dispatch Module	TC-REQ-04	A	Printable handover manifest generated
BBMS-F-020	Delivery confirmation	4.4	Dispatch / Custody	TC-REQ-05	A	Hospital receipt confirmation updates ledger
BBMS-F-021	Immutable audit trail	4.5	Audit / Security	TC-AUD-01	A	Append-only audit logs for all transactions
BBMS-F-022	Analytics dashboard	4.5	Admin / UI	TC-AUD-02	A	Real-time charts and KPI metrics verified
BBMS-F-023	CSV / PDF report export	4.5	Reporting Module	TC-AUD-03	A	Regulatory compliance reports exported
BBMS-F-024	Public availability portal	4.5	Public Portal	TC-AUD-04	A	Live stock counts visible without login
BBMS-F-025	Database backup & integrity	4.5	Database / Storage	TC-AUD-05	A	SQLite WAL integrity check passes

BBMS-NF-001	API response performance	5.0	API / Server	TC-NF-PERF-01	A	p95 latency < 250 ms verified
BBMS-NF-002	System availability & uptime	5.0	Core Architecture	TC-NF-REL-01	A	Zero crashes in continuous stress tests
BBMS-NF-003	UI responsiveness & WCAG	5.0	Frontend UI	TC-NF-UX-01	A	Mobile responsive; WCAG AA compliant
BBMS-NF-004	Cross-platform portability	5.0	Platform Runtime	TC-NF-PORT-01	A	Operates on macOS, Linux, and Windows
BBMS-NF-005	Modular architectural design	5.0	Codebase / Clean Arch	TC-NF-MNT-01	A	Layered architecture; zero lint errors
BBMS-SR-001	Salted bcrypt password hash	5.1	Security / Auth	TC-SEC-01	A	Zero plaintext passwords in storage/logs
BBMS-SR-002	Parameterized SQL queries	5.1	Security / DB	TC-SEC-02	A	Prepared statements prevent SQL injection
BBMS-SR-003	HttpOnly & SameSite cookies	5.1	Security / Session	TC-SEC-03	A	Cookie flags protect against XSS hijacking
BBMS-SR-004	RBAC route authorization	5.1	Security / Middleware	TC-SEC-04	A	Strict route access checks prevent escalation
BBMS-SR-005	Input validation & sanitization	5.1	Security / Input	TC-SEC-05	A	Strict schema validation on all endpoints
BBMS-SR-006	Rate limiting & brute force defense	5.1	Security / Network	TC-SEC-06	A	HTTP 429 triggered on rapid attempts